N83 D18: native validation completed

Run 2026-10-24 06:03:07–06:23:50 UTC · correctness and construction only

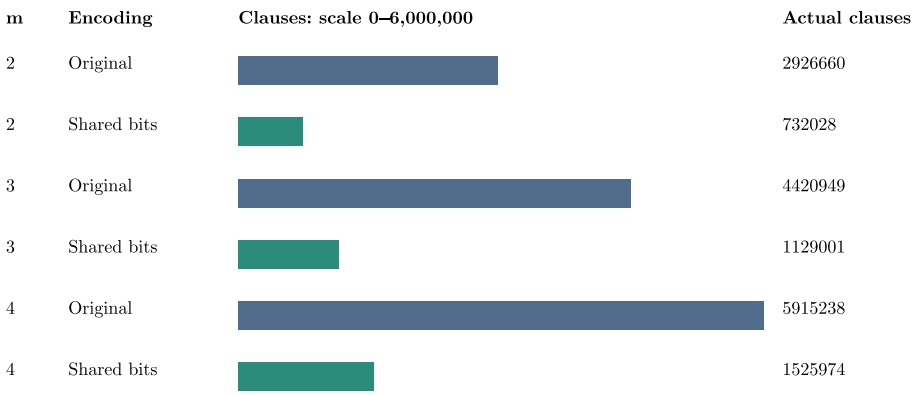
The requested local native gate passes on census parent `2e58bc5be` plus shared-bit implementation `4e1d311d8`. Encoding baseline: `e294f8c92`. The working curve is `icv1-f2m83-tm6151469093347-debefd74`.

Gate	Observed outcome
Full release library	4,478 passed; zero failed; 119 ignored
Eight affected examples	28 passed; zero failed
Release executables	Five builds, exit 0
Six full-size model constructions	All actual counts equal preflight
Receipt and source custody	Construction byte-identical; six source hashes match
Workflow metadata	440 workflows; zero budget violations; actionlint passes

D18: actual SAT construction counts

N83 · 16,627 fixed representatives · both encodings constructed for m=2,3,4

66 common root bits; accepted domain and choice wires unchanged.



m	Variables, both encodings	XOR rows, both encodings
2	90258	1743
3	145928	2822
4	201598	3901

One deterministic construction per cell. Exact integer counts; no sampling interval. Source: `construction.json`, native source digests recorded in that receipt. No solver search or runtime comparison is included in this construction check.

Apple M4 Pro / arm64, shared development host. Cargo records Rust 1.98.0 and LLVM 22.1.8. The busy lock serializes work; it does not qualify timing. There was no hard memory cap. Exact integer counts have no sampling interval. No SAT search or solver-runtime comparison was performed by this gate.

The earlier sandbox attempt passed 4,395 tests and failed nine local-socket tests with “Operation not permitted”; 118 were ignored. Its log is retained. The successful full suite above ran with socket access; no tests were disabled. The unmounted T7 run is not counted as a pass.

Remaining gates: bounded Linux construction; a committed actual-host card; 22 constructed controls, 22 independent replays, 18 natural queries; exact-head follow-up CI and merge. The Linux validation job is queued behind the benchmark lock. Natural useful rank, full cold/reused IC cost and matched rho remain open.

Evidence: `validation-v3.tar.xz` contains 34 files and was hash-checked after extraction and byte-identical repack. SHA-256: e6036d207cb7b2720d4cfd1c20eca943c8ba162224ad03e96087f5db31d3ab4c. Sources: `RUN_REPORT.md`, `validation-v3-files.sha256`, `construction.json`, `PROTOCOL.md`, and `PROOF_MAP.md`. The graph is unchanged because the new construction reproduces its source receipt exactly. Canonical runtime charts and benchmark rows stay unchanged: this run adds no timing comparison.